



October 11, 2019

Integra LifeSciences Corporation
Omayra Ramirez
Regulatory Affairs Specialist
311 Enterprise Drive
Plainsboro, New Jersey 08536

Re: K191684

Trade/Device Name: MoniTorr ICP External CSF Drainage and Monitoring System; LimiTorr Volume Limiting External CSF Drainage and Monitoring System

Regulation Number: 21 CFR 882.5550

Regulation Name: Central Nervous System Fluid Shunt and Components

Regulatory Class: Class II

Product Code: JXG

Dated: September 9, 2019

Received: September 10, 2019

Dear Omayra Ramirez:

We have reviewed your Section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database located at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's

requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR 803) for devices or postmarketing safety reporting (21 CFR 4, Subpart B) for combination products (see <https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR 1000-1050.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR Part 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems>.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance>) and CDRH Learn (<https://www.fda.gov/training-and-continuing-education/cdrh-learn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/contact-us-division-industry-and-consumer-education-dice>) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

Xiaolin Zheng, Ph.D., M.S.
Director (*Acting*)
DHT5A: Division of Neurosurgical,
Neurointerventional and Neurodiagnostic Devices
OHT5: Office of Neurological
and Physical Medicine Devices
Office of Product Evaluation and Quality
Center for Devices and Radiological Health

Enclosure

Indications for Use

510(k) Number (if known)

K191684

Device Name

MoniTor™ External CSF Drainage and Monitoring System

LimiTor™ Volume Limiting External CSF Drainage and Monitoring System

Indications for Use (Describe)

The MoniTor ICP™ system allows for drainage and monitoring of CSF from the lateral ventricles of the brain and the lumbar subarachnoid space in selected patients to reduce intracranial pressure (ICP), to monitor CSF, to provide temporary drainage of CSF in patients with infected CSF shunts, and the monitoring of ICP.

The LimiTor™ system allows for drainage and monitoring of CSF from the lateral ventricles of the brain and the lumbar subarachnoid space in selected patients to reduce intracranial pressure (ICP), to monitor CSF, to provide temporary drainage of CSF in patients with infected CSF shunts, and to monitor ICP.

Type of Use (Select one or both, as applicable)

☒ Prescription Use (Part 21 CFR 801 Subpart D)

☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

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510(k) SUMMARY

A summary of 510(k) safety and effectiveness information in accordance with the requirements of 21 CFR 807.92

807.92(a)(1) – Submitter information	
Name	Integra LifeSciences Corporation
Address	311 Enterprise Drive Plainsboro, NJ 08536 USA
Phone Number	1-609-275-0500
Establishment Registration Number	9004007
Name of Contact Person	Omayra Ramirez
Date Prepared	September 6, 2019
807.92(a)(2) – Name of device	
Trade or Propriety Name	LimiTorr™ Volume Limiting External CSF Drainage and Monitoring System MoniTorr ICP™ External CSF Drainage and Monitoring System
Common or Usual Name	External Cerebrospinal Fluid Drainage and Monitoring System
Classification Name	Central Nervous System Shunt and Components
Classification Panel	Neurology
Regulation	Class II, 21 CFR 882.5550
Product Code(s)	JXG
807.92(a)(3) - Legally marketed device(s) to which equivalence is claimed	
MoniTorr ICP™ External Drainage and Monitoring Systems; K920156 / K022554 LimiTorr™ Volume Limiting External CSF Drainage Systems; K072929	

807.92(a)(4) - Device description

MoniTorr:

The MoniTorr ICP™ External CSF Drainage and Monitoring Systems provide a closed system for the drainage of cerebrospinal fluid (CSF) from the ventricles of the brain or the lumbar subarachnoid space to a drainage bag. The system may be used with a pole mounted assembly that allows alignment with the patient and secure positive or negative pressure level setting. The system also has been designed to provide for ease of patient transport through a compact design and antimicrobial hydrophobic vent feature that resists occlusion.

LimiTorr:

The LimiTorr™ External CSF Drainage and Monitoring Systems provides a closed system for the CSF from the ventricles of the brain or the lumbar subarachnoid space to an external drainage bag. The LimiTorr volume limiting external CSF drainage system was designed to include a volume limiting valve mechanism which reduces the chances for excessive CSF drainage. The burette in the volume limiting external CSF drainage system contains a volume limiting valve which stops drainage when the pre-determined volume (20ml-30ml) is reached.

Both drainage systems are provided sterile and connect to either ventricular or lumbar catheters (not included in this submission) via a luer connection to a patient line and ultimately to a drainage bag. The indications for use of both MoniTorr and LimiTorr remain unchanged with this submission.

The intent of this 510(k) is to notify FDA of Integra's intent to implement a new stopcock component, make minor dimensional modifications to the stopcock manifold and to add a precaution to the Instruction for Use (IFU) concerning not over-tightening or applying excessive force to the luer lock/fluid filled transducer. The stopcocks control the flow of CSF to allow drainage from the patient, or allow flow to a pressure transducer to determine the intracranial pressure (ICP) of the patient. The stopcocks may be used to allow extraction of CSF from patient line for patient sampling. The proposed stopcock is a 4-way stopcock, with standard port connections, which are both characteristics of the predicate stopcock. The proposed stopcock maintains the same luer port internal diameter as the predicate.

807.92(a)(5) – Intended use of the device	
Indications for Use	The MoniTorr ICP™ system allows for drainage and monitoring of CSF from the lateral ventricles of the brain and the lumbar subarachnoid space in selected patients to reduce intracranial pressure (ICP), to monitor CSF, to provide temporary drainage of CSF in patients with infected CSF shunts, and the monitoring of ICP. The LimiTorr™ system allows for drainage and monitoring of CSF from the lateral ventricles of the brain and the lumbar subarachnoid space in selected patients to reduce intracranial pressure (ICP), to monitor CSF, to provide temporary drainage of CSF in patients with infected CSF shunts, and to monitor ICP.
807.92(a)(6) Summary of the technological characteristics of the device compared to the predicate	

The proposed MoniTorr ICP™ and LimiTorr™ External Drainage Systems with the new stopcock and additional precaution to the labeling have the same technological characteristics compared to the predicate devices. Other than the proposed stopcock and labeling change, there are no other changes to the MoniTorr or LimiTorr devices. Furthermore, the implementation of the new stopcock component to both systems and the addition of a new precaution to the labeling does not alter the indications for use, intended use, manufacturing and sterilization process, or the fundamental scientific technology of the devices. Both the predicate and proposed stopcocks use substantially equivalent materials (HDPE and polycarbonate) and the same lubricant. Table 1 provides a comparison of the predicate and proposed devices.

Table 1: Comparison of Subject and Predicate Devices

Property	Subject Devices	Predicates	Comments
Indications	The MoniTorr ICP™ system allows for drainage and monitoring of CSF from the lateral ventricles of the brain and the lumbar subarachnoid space in selected patients to reduce intracranial pressure (ICP), to monitor CSF, to provide temporary drainage of CSF in patients with infected CSF shunts, and the monitoring of ICP. The LimiTorr™ system allows for drainage and monitoring of CSF from the lateral ventricles of the brain and the lumbar subarachnoid space in selected patients to reduce intracranial pressure (ICP), to monitor CSF, to provide temporary drainage of CSF in patients with infected CSF shunts, and to monitor ICP.	Same	Same as predicate
Stopcock Materials	Handle Core: HDPE Body material: Polycarbonate Lubricant: Dow Medical Fluid	Handle Core: HDPE Body material: Polycarbonate Lubricant: Dow Medical Fluid	The predicate and proposed stopcock use the same base materials, including the same lubricant, only the viscosity is different. Biocompatibility testing demonstrates compliance to ISO 10993-1 and therefore the materials do not raise new questions of safety and effectiveness.

Stopcock Luer port Internal diameter	0.168 in	0.168 in	Same
Flow path (inner diameter) of the stopcock	Minimum 0.065 in (Nominal 0.085 in)	0.075 ± 0.005 in Nominal 0.075 in	Flow path specification is within specification of the predicate device. Performance testing supports that flow performance is not impacted by the new stopcock inner diameter. The minor differences in the inner diameter of the device flow path between the predicate and subject devices do not raise new questions of safety and effectiveness.
4-way stopcock (Yes/No)	Yes	Yes	Same
Standard female Luer Port (Yes/No)	Yes	Yes	Same
Manifold Bore dimension	0.343 ± 0.002 in	0.392 ± 0.001 in	A dimensional change to the manifold cavity where the stopcock is bonded was required because of the new stopcock. Performance testing and process verification supports that this dimensional change does not raise new questions of safety and effectiveness.
Sterility Information	Provide Sterile, single use. EO sterilized 10 -6	Provide Sterile, single use. EO sterilized 10 -6	Same as predicate
Packaging Configuration	Double Tyvek/ Mylar Pouch and white corrugated box	Double Tyvek/ Mylar Pouch and white corrugated box	Same as Predicate
807.92(b)(1-2) – Nonclinical and clinical tests submitted			

Non-clinical testing was performed to ensure the safety and efficacy of the modified LimiTorr and MoniTorr External Drainage and Monitoring Systems. Testing included:

Table 2: Performance Bench Test Results		
Test	Test Summary	Results
Torque to Failure	To determine if the average torque to failure for the stopcock for LimiTorr and MoniTorr ICP are within current specifications.	All results passed, demonstrating substantial equivalence of the proposed device.
Average Peak Moment	To determine if the average peak moment of the LimiTorr and MoniTorr ICP stopcock is within current specifications.	All results passed, demonstrating substantial equivalence of the proposed device.
Peak Tensile Force to overcome bond strength	To determine if the average tensile force required to remove the bonded tubing from the stopcock is within current specifications.	All results passed, demonstrating substantial equivalence of the proposed device.
Resistance to Flow	To determine if the resistance to flow of the device is within current specifications.	All results passed, demonstrating substantial equivalence of the proposed device.
Leakage Test	To determine if the pressure decay in the patient line is within current specifications.	All results passed, demonstrating substantial equivalence of the proposed device.
30 Day functionality test for LimiTorr and MoniTorr	To determine if the stopcock/tubing assembly for LimiTorr and MoniTorr ICP continue to meet current specifications for torque to failure, peak moment, and peak tensile force to overcome bond strength after 30 days of simulated use.	All results passed, demonstrating substantial equivalence of the proposed device.
Design Validation	To confirm that the MoniTorr ICP and LimiTorr External CSF Drainage Systems meet the user needs, and that the system can be operated as intended, safely and effectively.	All results passed, demonstrating substantial equivalence of the proposed device.

In addition to bench testing, biocompatibility testing was leverage from the stopcock vendor. The device tested (Safeport with Handle) is identical to the proposed stopcock having the same chemical formulations, same surface properties, same manufacturing, and same sterilization processes. Furthermore, there are no additional materials added to the stopcock during Integra's manufacturing process. The biocompatibility testing is summarized in Table 3.

Table 3: Biocompatibility Test Summary		
Test	Results	Conclusions
Cytotoxicity ISO Elution Method	No cytotoxicity or cell lysis in any of the test wells	Non-cytotoxic
ISO Guinea Pig Maximization Sensitization	Test article extracts showed no evidence of causing delayed dermal contact sensitization in the guinea pig.	Non-sensitizing
ISO Intracutaneous Study in Rabbits	The test article met the requirements of the test since the difference between each test article extract overall mean score and corresponding control extract overall mean score was 0.0 and 0.1 for the SC and SO test article extracts, respectively.	Non-irritating
ISO Acute Systemic Toxicity Study in Mice	- No mortality during the study - All animals were clinically normal throughout the study - Body weight data were acceptable	Non-toxic acutely
USP Rabbit Material-Mediated Pyrogenicity	The test article met the requirements for the absence of pyrogens. No single animal showed a temperature rise of 0.5°C or more above its baseline temperature. The total rise of the rabbits' temperature during the 3 hour observation period was 0.2°C.	Non-pyrogenic
ASTM Hemolysis	The hemolytic index for the test article in direct contact with blood was 0.0%, and the hemolytic index for the test article extract was 0.0%.	Non-hemolytic

Testing was successfully completed for all protocols above and supports the conclusion that all product specifications and design inputs have been met. Integra LifeSciences therefore believes verification and validation testing results for the proposed MoniTorr and LimiTorr support a determination of substantial equivalence when compared to the predicate devices.

No clinical studies were performed or required as all conducted performance tests appropriately support a

determination of substantial equivalence compared to the predicate devices.
807.92(b)(3) – Conclusions drawn from non-clinical and clinical data
The results of the non-clinical testing indicate that the intended use of the device, fundamental scientific technology, and performance of the MoniTorr and LimiTorr are substantially equivalent to the predicate devices.